

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Sony
Firmware Version: N/A
Model(s): VPL-FHZ70, VPL-FHZ75

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.04.0001-b002	2.3.1

Version History

Module Version	Date	Notes
1_0_0_0	9/19/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='VPL-FHZ70')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='VPL-FHZ70')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 53595, Model='VPL-FHZ70')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio ^{1,2,3}	Value '4:3' 'Full 2' 'Zoom'	Value '16:9' 'Normal'	Value 'Full 1' 'Full'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage ^{1,4}	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command ClosedCaption	Value 'Off' 'CC3' 'Text2'	Value 'CC1' 'CC4' 'Text3'	Value 'CC2' 'Text1' 'Text4'
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'Off')			
Command ExecutiveMode	Value 'Off'	Value 'On'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'Off')			
Command Focus ⁹	Value 'Far'	Value 'Near'	Value 'Focus'
# Focus example InterfaceName.Set('Focus', 'Far')			
Command Freeze ^{1,2,4,5}	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input ⁶	Value 'Video' 'HDMI'	Value 'RGB' 'HDBaseT'	Value 'DVI'
# Input example InterfaceName.Set('Input', 'Video')			
Command LampMode	Value 'Standard' 'Extended'	Value 'Middle'	Value 'Custom'
# LampMode example InterfaceName.Set('LampMode', 'Standard')			
Command LensControl	Value 'On'	Value 'Off'	
# LensControl example InterfaceName.Set('LensControl', 'On')			

Command LensShift ⁹	Value 'Up' 'Right'	Value 'Down' 'Shift'	Value 'Left'
# LensShift example InterfaceName.Set('LensShift', 'Up')			
Command MenuNavigation	Value 'Menu' 'Right' 'Return'	Value 'Up' 'Left'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command MultiScreen ⁷	Value 'On'	Value 'Off'	
# MultiScreen example InterfaceName.Set('MultiScreen', 'On')			
Command PictureMode ^{2,8}	Value 'Dynamic' 'Multi Screen'	Value 'Standard'	Value 'Brightness Priority'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			
Command Zoom ⁹	Value 'Up'	Value 'Down'	Value 'Zoom'
# Zoom example InterfaceName.Set('Zoom', 'Up')			

¹ Unavailable when there is no signal.

² Unavailable when Multi Screen is On.

³ Availability depends on the current Input and the Screen Aspect menu option.

⁴ Available when the Input is RGB and depends on the input signal type.

⁵ Available when the Input is DVI.

⁶ When Multi Screen is On, only RGB, DVI, and HDMI are available.

⁷ Not available when the Screen Aspect menu option is 4:3.

⁸ Unavailable when the Intelligent Setting menu option is On.

⁹ Available when Lens Control is On.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio ^{1, 2, 3}	Value '4:3' 'Full 2' 'Zoom'	Value '16:9' 'Normal'	Value 'Full 1' 'Full'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ClosedCaption	Value 'Off' 'CC3' 'Text2'	Value 'CC1' 'CC4' 'Text3'	Value 'CC2' 'Text1' 'Text4'
# ClosedCaption example InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'No Error' 'System Error 2' 'Shock Error' 'Temperature Error' 'Luminance Error'	Value 'Power Supply Error' 'Cover Error' 'Lens Not Attached Error' 'Fan Error' 'Assembling Error'	Value 'Power Supply (D5V) Error' 'Light-source Error' 'Installation Angle Error' 'Wheel Rotation Error' 'Lens Shift Error'
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus')			

Global Scripter Module Communication Sheet

InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'Off'	Value 'On'	
# ExecutiveMode example InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command Freeze ^{1, 2, 4, 5}	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input ⁶	Value 'Video' 'HDMI'	Value 'RGB' 'HDBaseT'	Value 'DVI'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampHours	Value Hours		
# LampHours example InterfaceName.Update('LampHours') Value = InterfaceName.ReadStatus('LampHours') InterfaceName.SubscribeStatus('LampHours', None, FeedbackHandler)			
Command LampMode	Value 'Standard' 'Extended'	Value 'Middle'	Value 'Custom'
# LampMode example InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LensControl	Value 'On'	Value 'Off'	
# LensControl example InterfaceName.Update('LensControl') Value = InterfaceName.ReadStatus('LensControl') InterfaceName.SubscribeStatus('LensControl', None, FeedbackHandler)			
Command MultiScreen ⁷	Value 'On'	Value 'Off'	
# MultiScreen example InterfaceName.Update('MultiScreen') Value = InterfaceName.ReadStatus('MultiScreen') InterfaceName.SubscribeStatus('MultiScreen', None, FeedbackHandler)			
Command OperationHours	Value Hours		
# OperationHours example InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)			
Command PictureMode ^{2, 8}	Value 'Dynamic' 'Multi Screen'	Value 'Standard'	Value 'Brightness Priority'
# PictureMode example InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode')			

InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command	Value	Value	Value
Power	'On'	'Startup'	'Cooling 1'
	'Cooling 2'	'Power Saving Cooling 1'	'Power Saving Cooling 2'
	'Power Saving Standby'	'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command	Value		
Volume	0 to 100 in steps of 1		
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

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² Unavailable when Multi Screen is On.

³ Availability depends on the current Input and the Screen Aspect menu option.

⁴ Available when the Input is RGB and depends on the input signal type.

⁵ Available when the Input is DVI.

⁶ When Multi Screen is On, only RGB, DVI, and HDMI are available.

⁷ Not available when the Screen Aspect menu option is 4:3.

⁸ Unavailable when the Intelligent Setting menu option is On.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Ensure that the follow setting in the projector's OSD menu is set for the driver to work properly, depending if you are using RS-232C or HDBaseT communication:

- Connection/Power Menu → HDBaseT Settings → RS-232C Setting → **RS-232C**
- Connection/Power Menu → HDBaseT Settings → RS-232C Setting → **via HDBaseT**

The baud rate is different depending on which communication method you select. When connecting via HDBaseT, the connection is at 9,600 bps. When using the RS-232C Terminal of the main unit, you can connect at 38,400 bps.

Serial communication

Port Type: RS-232

Baud Rate: 38400

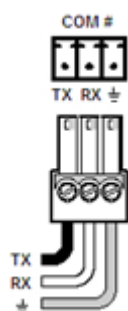
Data Bits: 8

Parity: Even

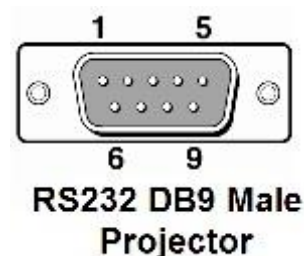
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 53595

Logon Credentials Supported: Yes

Default Password: Projector

Multi-Connection Yes

Capabilities:

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Ensure that the follow settings in the projector are set for the driver to work properly:

1. Access projector's Web Control web page by typing in the projector's IP address in a web browser and enable Setup → Advanced Menu → ADCP → **Start ADCP Service**.
2. On the same page, enable **Requires Authentication** if you require a password. If you don't require a password, disable this setting. Note that the password is the same as the administrator password required when gaining access to the Setup page in Web Control.
3. Set the following settings in the projector's OSD menu:
 - Connection/Power Menu → HDBaseT Settings → LAN Setting → **LAN Port**
 - Connection/Power Menu → ECO → Standby Mode → **Standard**

Appendix A. Set Commands

Aspect Ratio 16:9	aspect "16_9"\x0D\x0A
Aspect Ratio 4:3	aspect "4_3"\x0D\x0A
Aspect Ratio Full	aspect "full"\x0D\x0A
Aspect Ratio Full 1	aspect "full1"\x0D\x0A
Aspect Ratio Full 2	aspect "full2"\x0D\x0A
Aspect Ratio Normal	aspect "normal"\x0D\x0A
Aspect Ratio Zoom	aspect "zoom"\x0D\x0A
Audio Mute Off	muting "off"\x0D\x0A
Audio Mute On	muting "on"\x0D\x0A
Auto Image None	apa_exec\x0D\x0A
Closed Caption CC1	cc_display "cc1"\x0D\x0A
Closed Caption CC2	cc_display "cc2"\x0D\x0A
Closed Caption CC3	cc_display "cc3"\x0D\x0A
Closed Caption CC4	cc_display "cc4"\x0D\x0A
Closed Caption Off	cc_display "off"\x0D\x0A
Closed Caption Text1	cc_display "text1"\x0D\x0A
Closed Caption Text2	cc_display "text2"\x0D\x0A
Closed Caption Text3	cc_display "text3"\x0D\x0A
Closed Caption Text4	cc_display "text4"\x0D\x0A
Executive Mode Off	controlkey_lock "off"\x0D\x0A
Executive Mode On	controlkey_lock "on"\x0D\x0A
Focus Far	key "lens_focus_far"\x0D\x0A
Focus Focus	key "lens_focus"\x0D\x0A
Focus Near	key "lens_focus_near"\x0D\x0A
Freeze Off	freeze "off"\x0D\x0A
Freeze On	freeze "on"\x0D\x0A
Input DVI	input "dvi1"\x0D\x0A
Input HDBaseT	input "hdbaset1"\x0D\x0A
Input HDMI	input "hdm1"\x0D\x0A
Input RGB	input "rgb1"\x0D\x0A
Input Video	input "video1"\x0D\x0A
Lamp Mode Custom	light_output_mode "custom"\x0D\x0A
Lamp Mode Extended	light_output_mode "extended"\x0D\x0A
Lamp Mode Middle	light_output_mode "mid"\x0D\x0A
Lamp Mode Standard	light_output_mode "high"\x0D\x0A
Lens Control Off	lens_lock "off"\x0D\x0A
Lens Control On	lens_lock "on"\x0D\x0A
Lens Shift Down	key "lens_shift_down"\x0D\x0A
Lens Shift Left	key "lens_shift_left"\x0D\x0A
Lens Shift Right	key "lens_shift_right"\x0D\x0A
Lens Shift Shift	key "lens_shift"\x0D\x0A
Lens Shift Up	key "lens_shift_up"\x0D\x0A

Global Scriptor Module Communication Sheet

Menu Navigation Down	key "down"\x0D\x0A
Menu Navigation Enter	key "enter"\x0D\x0A
Menu Navigation Left	key "left"\x0D\x0A
Menu Navigation Menu	key "menu"\x0D\x0A
Menu Navigation Return	key "return"\x0D\x0A
Menu Navigation Right	key "right"\x0D\x0A
Menu Navigation Up	key "up"\x0D\x0A
Multi Screen Off	multi_screen "off"\x0D\x0A
Multi Screen On	multi_screen "on"\x0D\x0A
Picture Mode Brightness Priority	picture_mode "brt_priority"\x0D\x0A
Picture Mode Dynamic	picture_mode "dynamic"\x0D\x0A
Picture Mode Multi Screen	picture_mode "multi_screen"\x0D\x0A
Picture Mode Standard	picture_mode "standard"\x0D\x0A
Volume 0	volume 0\x0D\x0A
Volume 100	volume 100\x0D\x0A
Zoom Down	key "lens_zoom_down"\x0D\x0A
Zoom Up	key "lens_zoom_up"\x0D\x0A
Zoom Zoom	key "lens_zoom"\x0D\x0A

Appendix B. Update Commands

Aspect Ratio	aspect ?\x0D\x0A
Audio Mute	muting ?\x0D\x0A
Closed Caption	cc_display ?\x0D\x0A
Device Status	error ?\x0D\x0A
Executive Mode	controlkey_lock ?\x0D\x0A
Freeze	freeze ?\x0D\x0A
Input	input ?\x0D\x0A
Lamp Hours	timer ?\x0D\x0A
Lamp Mode	light_output_mode ?\x0D\x0A
Lens Control	lens_lock ?\x0D\x0A
Multi Screen	multi_screen ?\x0D\x0A
Picture Mode	picture_mode ?\x0D\x0A
Power	power_status ?\x0D\x0A
Volume	volume ?\x0D\x0A